



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 14: Profit Maximization

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ICS2523 Microeconomics

Lecture 14

Last lecture we studied

- Firm's expansion path
- Cost functions
- Economies of Scale

In today's lecture, we will learn more about
costs in the short- and long-run

- Distinguish the Short- and Long-run
- Learn about the non-optimality of the short-run
- Understand the relationship of the cost curves in the short and long-run

Class assistantship today

Please remember that today we will have a face-to-face class assistantship at 14.50 at B22.

Homework 1 Lecture 13

Imagine your firm faces a short-run cost function equal to:

$$C_{SR} = \bar{k}^2 + 16 \frac{q^2}{\bar{k}}$$

- a. Find AC, MC, AVC in the short-run and graph them
- b. Find the Total Cost, AC, and MC curves in the long-run. Graph them.

HINT:

For b. remember that the total cost curve (or average cost curve) in the long run is the envelope of the total cost curve (or average cost curve) in the short run.

Homework L13: Solution

Imagine your firm faces a short-run cost function equal to:

$$C_{SR} = \bar{k}^2 + 16 \frac{q^2}{\bar{k}}$$

- a. Find AC, MC, AVC in the short-run and graph them

$$AC_{SR} = \frac{C_{SR}}{q} = \frac{\bar{k}^2}{q} + 16 \frac{q}{\bar{k}}$$

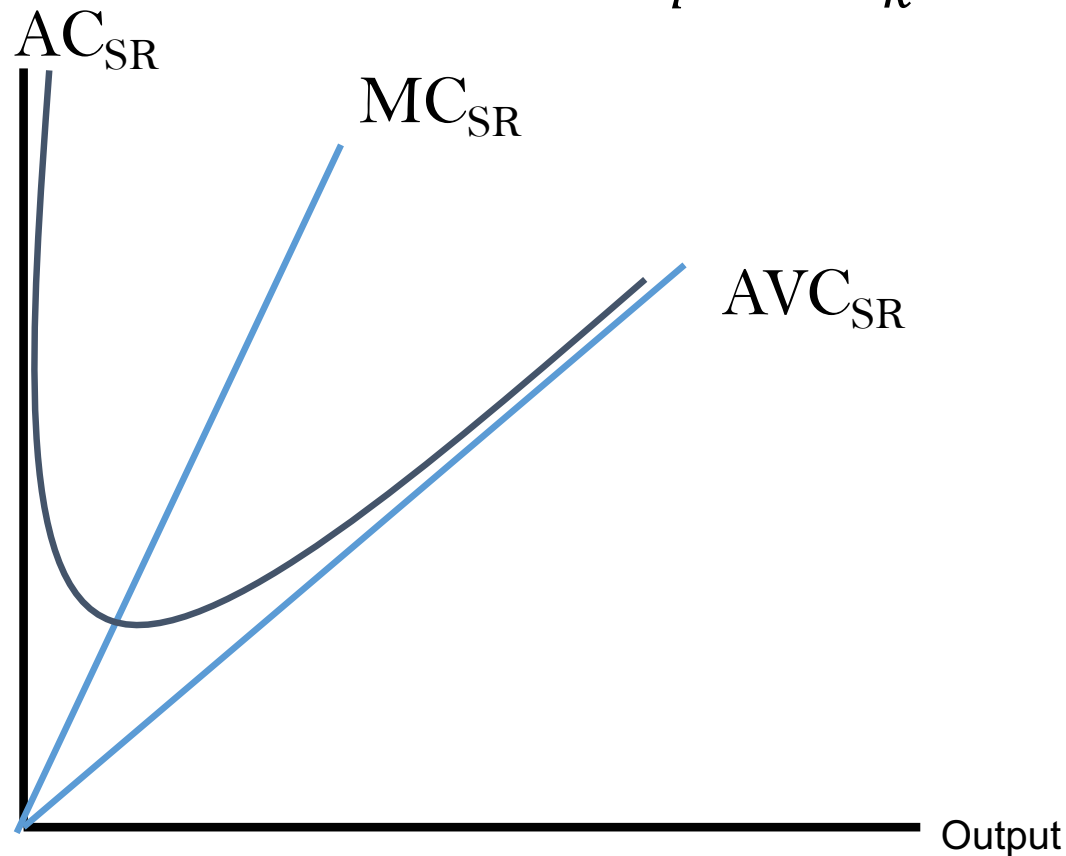
$$MC_{SR} = \frac{\partial C_{SR}}{\partial Q} = 32 \frac{q}{\bar{k}}$$

$$AVC_{SR} = \frac{VC_{SR}}{q} = 16 \frac{q}{\bar{k}}$$

Homework L13: Solution

$$AC_{SR} = \frac{C_{SR}}{q} = \frac{\bar{k}^{-2}}{q} + 16 \frac{q}{\bar{k}}, MC_{SR} = \frac{\partial C_{SR}}{\partial q} = 32 \frac{q}{\bar{k}}$$

$$AVC_{SR} = \frac{VC_{SR}}{q} = 16 \frac{q}{\bar{k}}$$



Homework L13: Solution

- b. Find the Total Cost, AC, and MC curves in the long-run. Graph them.

As seen in L13, the long-run total cost curve (or the Average Cost curve) must obey the short-run relationship + the condition that k must be cost minimizing for any output level. A first-order condition for this minimization is that:

$$\frac{\partial C_{SR}(r, w, q, k_1)}{\partial k_1} = 0$$

Homework L13: Solution

$$\frac{\partial C_{SR}(r, w, q, \bar{k})}{\partial \bar{k}} = 0$$

$$\frac{\partial (\bar{k}^2 + 16 \frac{q^2}{\bar{k}})}{\partial \bar{k}} = 0$$

$$2\bar{k} - 16 \frac{q^2}{\bar{k}^2} = 0$$

$$\bar{k}^3 = 8q^2$$

$$\bar{k} = 2q^{2/3}$$

Homework L13: Solution

$$\bar{k} = 2q^{2/3}$$

Replacing this value in the total cost curve of the short-run:

$$C_{SR} = \bar{k}^2 + 16 \frac{q^2}{\bar{k}}$$

$$C_{LR} = 4q^{4/3} + 16 \frac{q^2}{2q^{2/3}} = 4q^{4/3} + 8q^{4/3}$$

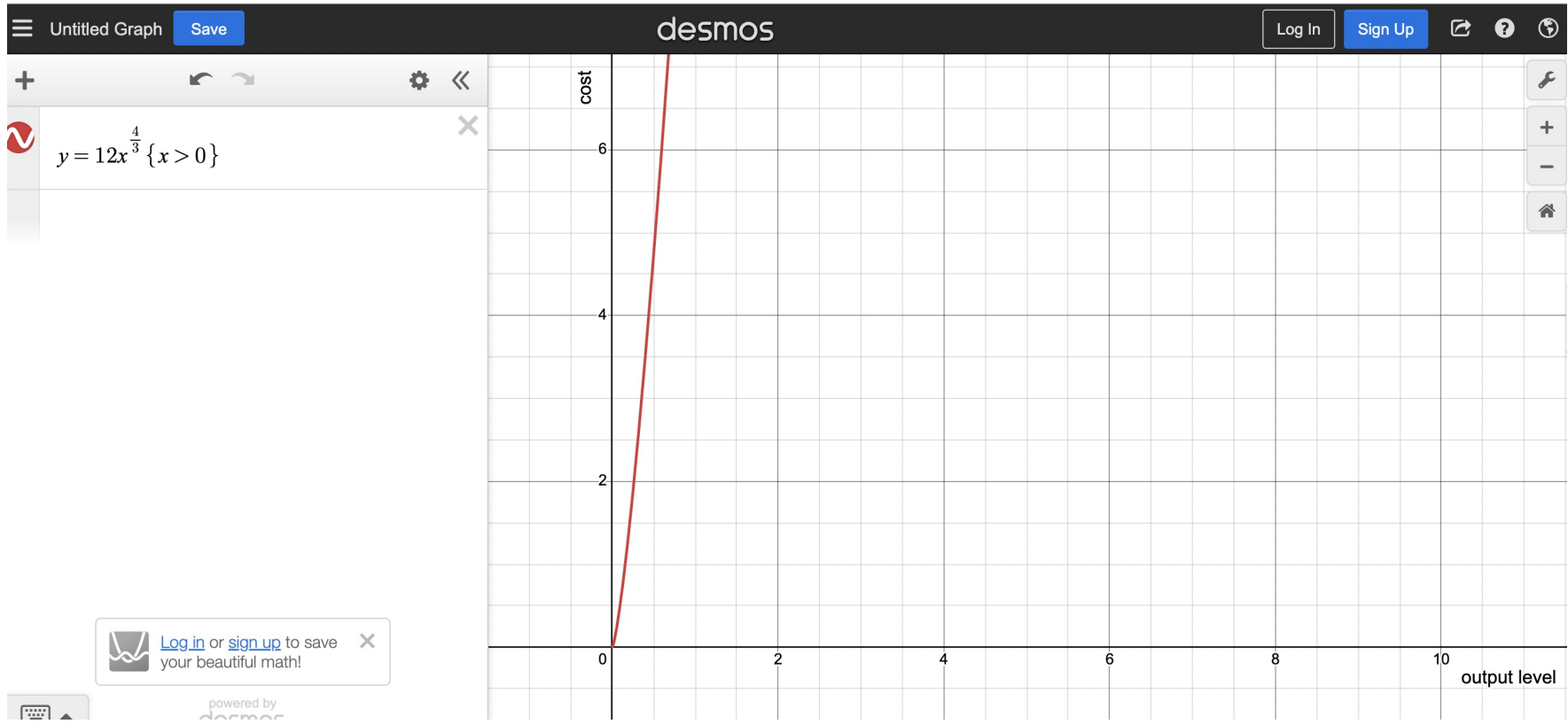
$$C_{LR} = 12q^{4/3}$$

$$AC_{LR} = 12q^{1/3}$$

$$MC_{LR} = 4q^{1/3}$$

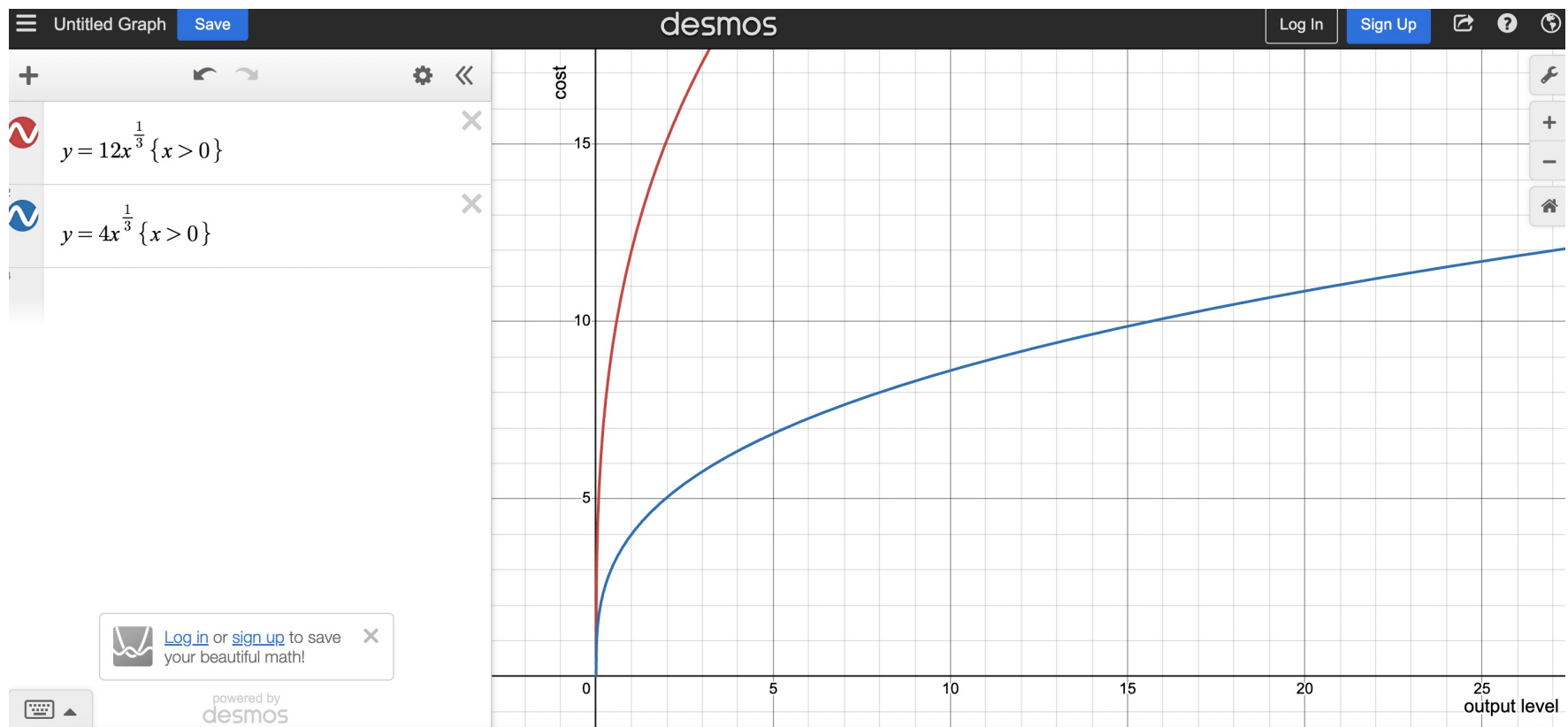
Homework L13: Solution

$$C_{LR} = 12q^{4/3}$$



Homework L13: Solution

$$AC_{LR} = 12q^{1/3}$$
$$MC_{LR} = 4q^{1/3}$$



Homework 2 Lecture 13

A firm has the fixed proportion production function $q = \min(2l, 3k)$, where k is capital and l is labour. The prices are r and w , respectively.

In the short-run, k is fixed at 2.

Graph: Total Cost, Average Cost and Marginal Cost in the **short-run** vs Average Cost and Marginal Cost in the **long-run**

Homework 2 L13: Solution

Short-run

$q = \min(2l, 3k)$, where k is capital and l is labour. The prices are r and w , respectively.

In the short-run, k is fixed at 2.

If k is fixed $\rightarrow 3k$ is also fixed and equal to six

If we graph $k = 2$, we can find the number of labour hours needed to reach different levels of output.

Homework 2 L13: Solution

If we graph $k = 2$, we can find the number of labour hours needed to reach different levels of output.



Homework 2 L13: Solution

Also, we can see that there are unreachable output levels at $k = 2$. Since $3k = 6$, the maximum level of output is 6.

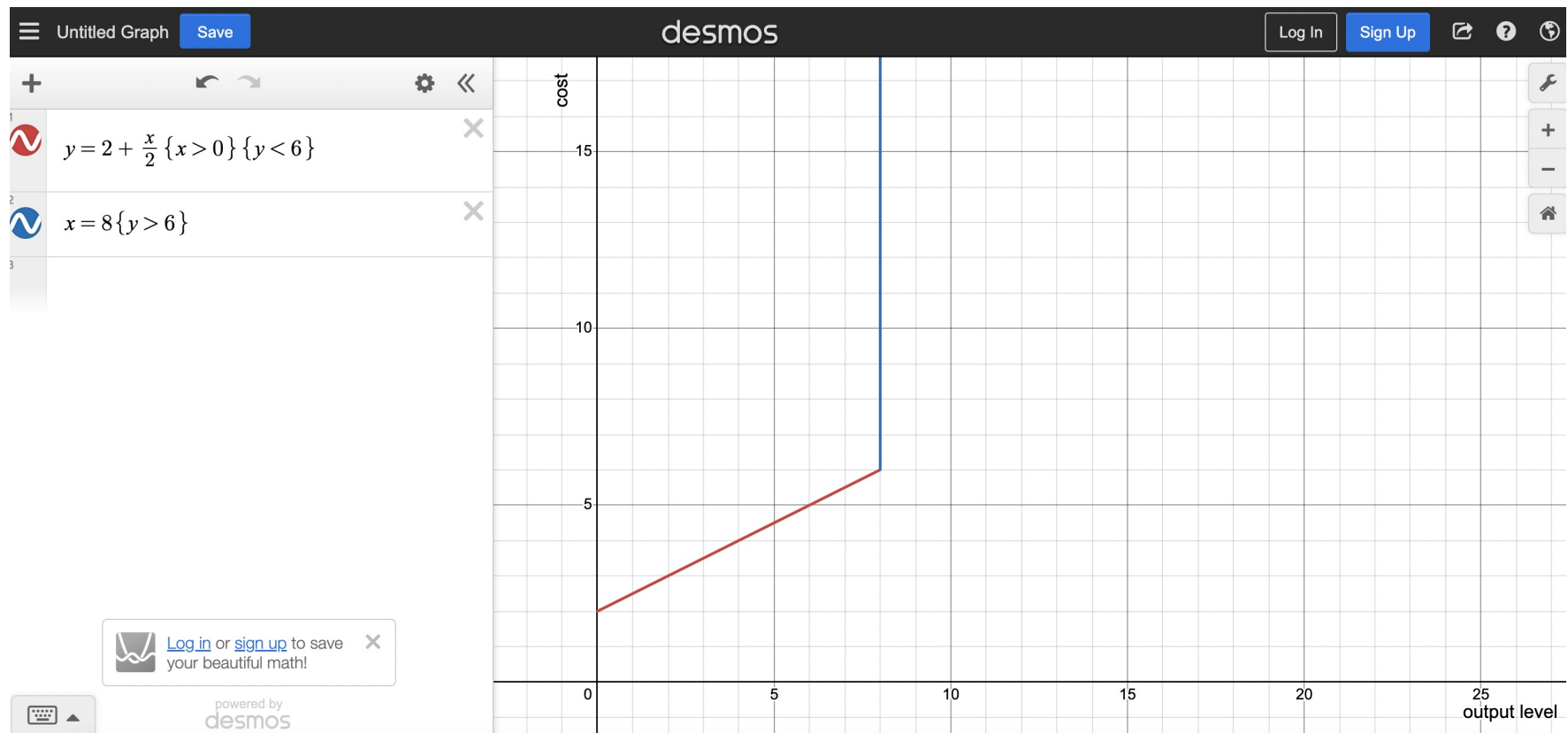
For output levels lower or equal to 6, the level of labour needed is: $2l = q \rightarrow l = q/2$

Thus, the total cost (for output levels lower or equal to 6) is:

$$C_{SR} = rk + wl = 2r + \frac{wq}{2}$$

Homework 2 L13: Solution

$$C_{SR} = \begin{cases} rk + wl = 2r + \frac{wq}{2}, & \text{if } 0 \leq q < 6 \\ \infty, & \text{if } q \geq 6 \end{cases}$$



Homework 2 L13: Solution

Average and Marginal cost in the short-run only exist if $q < 6$.

$$C_{SR} = rk + wl = 2r + \frac{wq}{2}$$

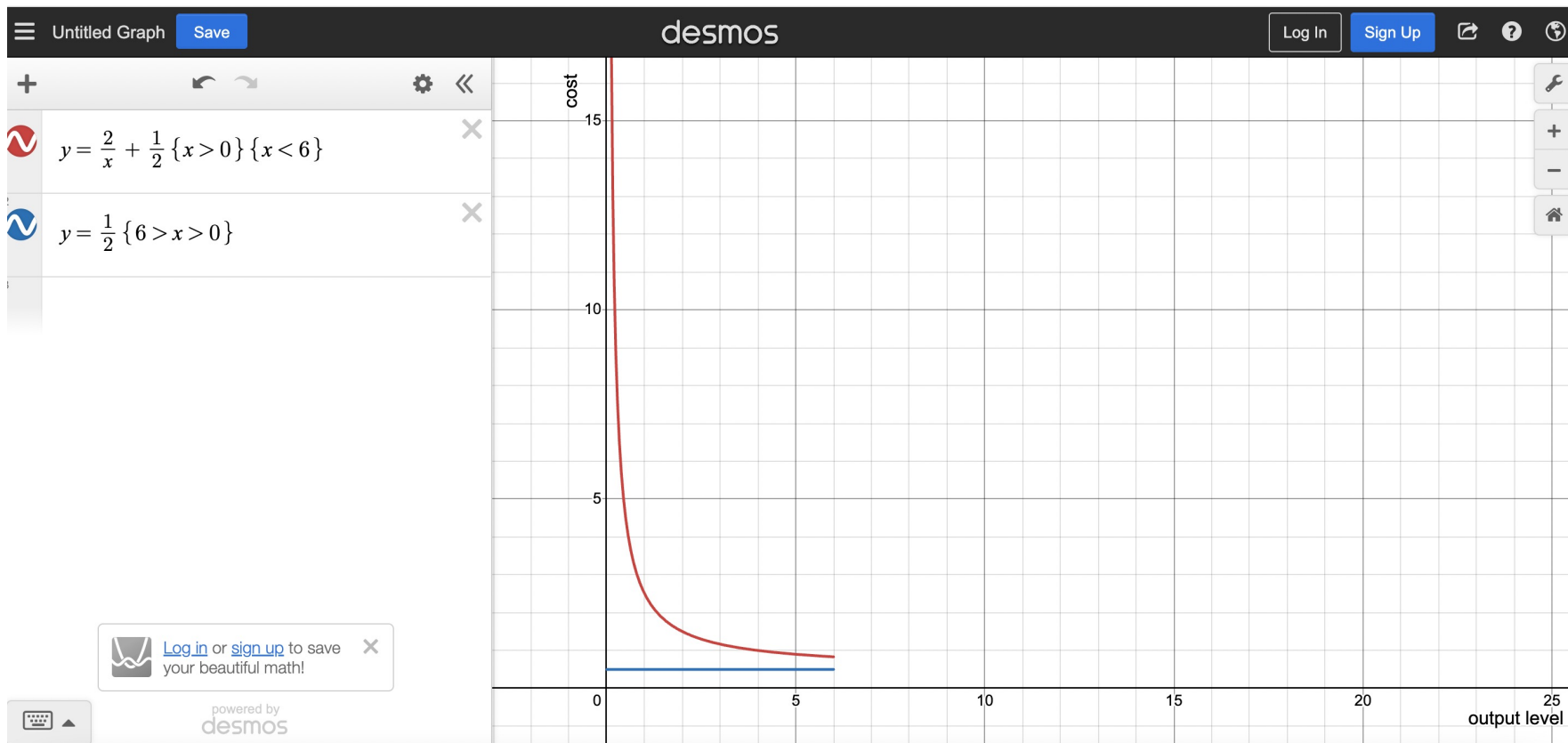
$$AC_{SR} = \frac{2r}{q} + \frac{w}{2}$$

$$AVC_{SR} = \frac{w}{2}$$

$$MC_{SR} = \frac{w}{2}$$

Homework 2 L13: Solution

$$AC_{SR} = \frac{2r}{q} + \frac{w}{2}, \quad AVC_{SR} = \frac{w}{2}, \quad MC_{SR} = \frac{w}{2}$$



Homework 2 L13: Solution

Long-run

$q = \min(2l, 3k)$, where k is capital and l is labour. The prices are r and w , respectively.

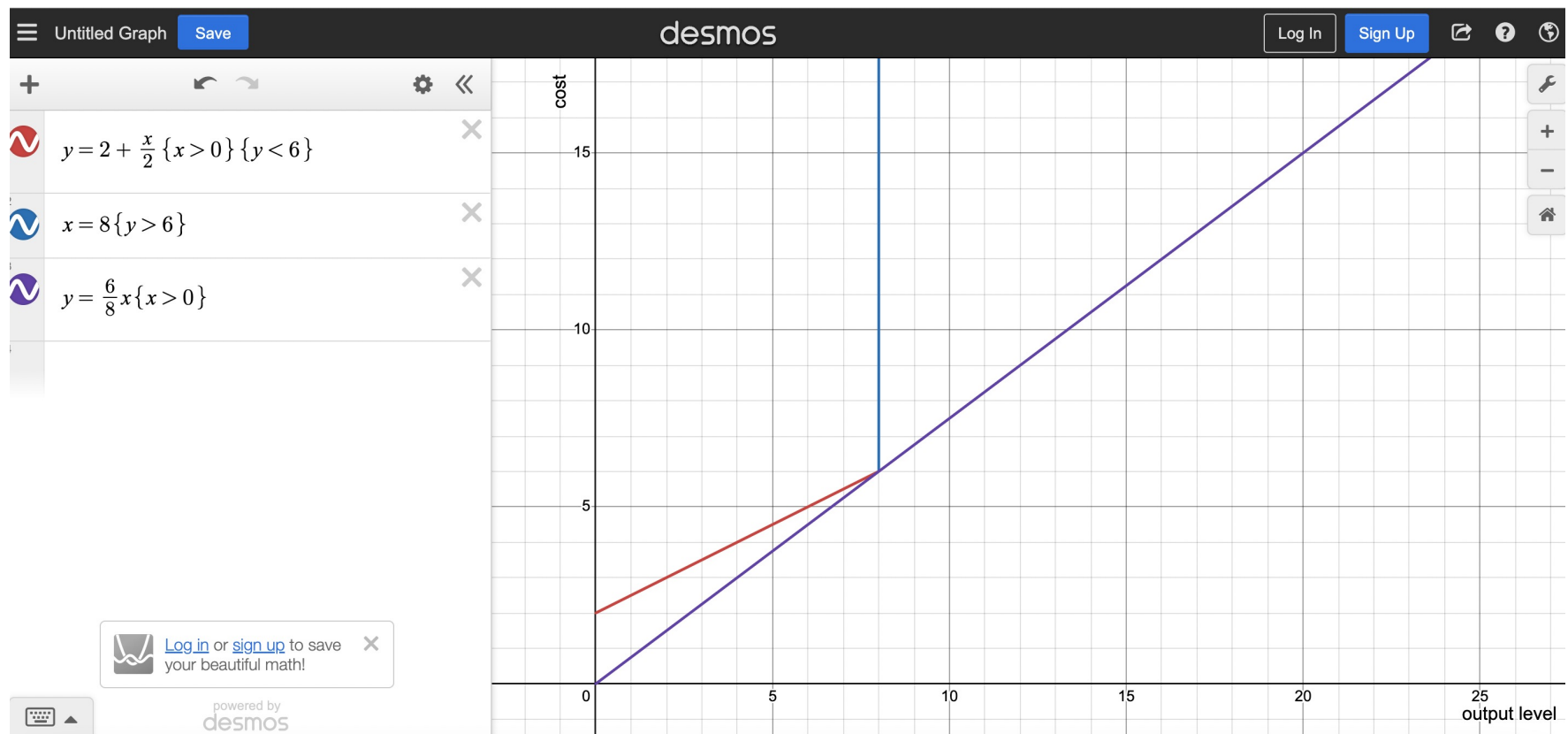
In the long-run, k is not fixed anymore. Thus, we can minimize costs $\rightarrow 2l = 3k = q \rightarrow l^* = \frac{q}{2}, k^* = \frac{q}{3}$

$$C_{LR} = rk^* + wl^* = r\frac{q}{3} + w\frac{q}{2} = \text{constant} \cdot q$$

$$AC_{LR} = MC_{LR} = \frac{r}{3} + \frac{w}{2} = \text{constant}$$

Homework 2 L13: Solution

$$C_{LR} = rk^* + wl^* = r \frac{q}{3} + w \frac{q}{2} = \text{constant} \cdot q$$



Profit Maximization

Producer Theory

14. Profit Maximization

Why are we studying all this?

Our ultimate goal is to model the producer's behavior →

Maximizing profits

How can we write a mathematical equation of profit?

Profit = Revenue - Costs

→ In order to **model the producer's profit**, we had to define the costs

→ To **model costs**, we need to know how the output is produced → production function

14. Profit Maximization

Steps into finding producer's profits:

- ① Have a production function $\rightarrow$ isoquant (output held constant)
- ② Know the input's prices
- ③ Find the cost function $\rightarrow$ minimizing the cost input choices (for a certain level of production)

WARNING! Long-run versus short-run

- ④ Only when we have the cost function we can **maximize profits (choosing a level of output)** , and we will assume the firm is a price taker

14. Profit Maximization

A **profit-maximizing firm** will choose its inputs and output to achieve maximum economic profits.

If a firm sells some level of output (q) at a market price of p per unit, the total revenues (R) are given by:

$$R(q) = p(q) \cdot q$$

By producing q , the firm incurs in some costs ($C(q)$)

14. Profit Maximization

The economic profits (π) will be:

Profits = Revenue - Costs

$$\pi(q) = p(q) q - C(q) = R(q) - C(q)$$

First order condition:

$$\frac{d\pi}{dq} = \frac{dR}{dq} - \frac{dC}{dq} = 0$$

$$\frac{dR}{dq} = \frac{dC}{dq} \rightarrow MR = MC$$

Second order condition: $\left. \frac{d^2 \pi}{dq^2} \right|_{q=q^*} < 0$

Marginal Revenue

Producer Theory

15. Marginal Revenue

What is the Marginal Revenue?

Revenue obtained from selling one more unit of output

$$MR(q) = \frac{dR}{dq} = \frac{d(p(q) \cdot q)}{dq} = p + q \frac{dp}{dq}$$

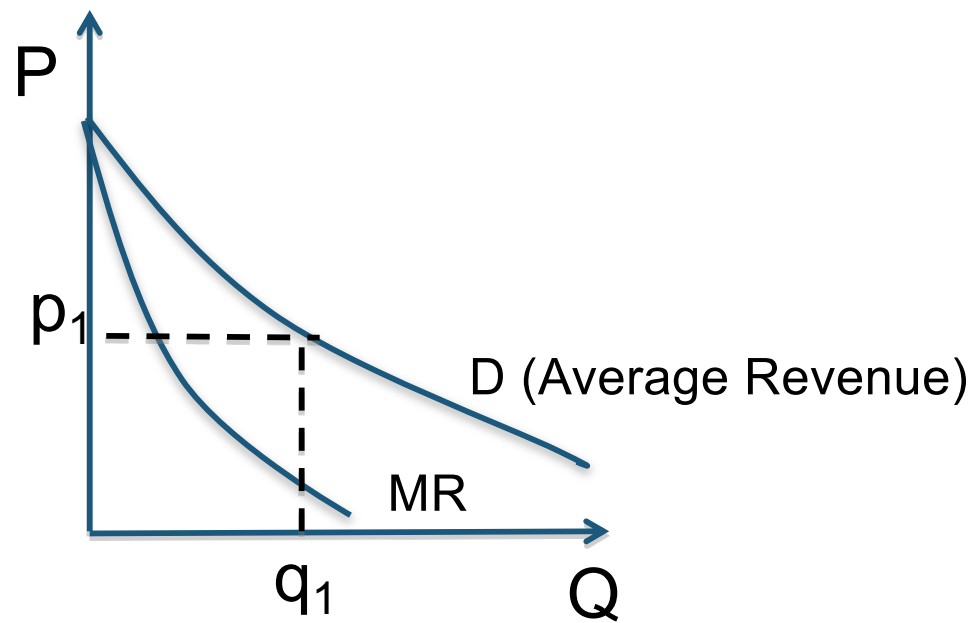
- If a firm's output decisions will not affect the market price $\frac{\partial p}{\partial q} = 0 \rightarrow MR=p$ and the **firm is a price taker**

15. Marginal Revenue

Marginal Revenue Curve:

Any demand curve has a **MR curve** associated with it

If the firm must sell all its output at one price (common assumption), we can think of the Demand curve as an Average Revenue Curve



15. Marginal Revenue

- When a firm is in a competitive market $\rightarrow$ price-taker
 $\rightarrow$ Price is FIXED!
- QUESTION: What happens to Marginal Revenue when a firm is in a competitive market?

$$MR = P^*$$

$\rightarrow$ First order condition is now $P^* = MC$

$$\frac{d\pi}{dq} = \frac{dR}{dq} - \frac{dC}{dq} = 0$$

$$\frac{d\pi}{dq} = \frac{d(P \cdot q)}{dq} - \frac{dC}{dq} = 0$$

$$\frac{d\pi}{dq} = P + q \frac{dP(q)}{dq} - \frac{dC}{dq} = 0$$

$P = MC$

15. Marginal Revenue

Second order condition (for maximizing profits):

$$\left. \frac{d^2 \pi}{dq^2} \right|_{q=q^*} < 0$$

$$\left. \frac{\partial (\cancel{P}^{*0} - MC)}{\partial q} \right|_{q=q^*} < 0$$

$$-\frac{\partial MC}{\partial q} < 0$$

$$\frac{\partial MC}{\partial q} > 0 \quad \rightarrow \text{Marginal Costs must be increasing if we are maximizing profits}$$

Supply curve in the short-run

Producer Theory

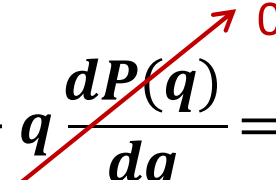
16. Supply Curve in the SR

Short-run Supply Curve by a Price-taking Firm:

We are going to study the **supply decision** of a price-taking firm.

The firm's short-run supply curve shows how much it will produce at various possible output prices.

To maximize its profit, the firm maximizes its revenue minus its costs::

$$MR(q) = P + q \frac{dP(q)}{dq} = MC(q)$$


If the firm is a price taker, the price P is given $\rightarrow MC = P^*$

So, the **supply curve should be the marginal cost curve**...but should it be over its entire interval?

16. Supply Curve in the SR

Short-run Supply Curve by a Price-taking Firm:

If the company is a price taker:

- a) When maximizing profit, the first order condition is: $MR = MC \rightarrow P = MC$
- b) The second order condition is $\rightarrow \partial MC / \partial q > 0$
*Therefore, the **Marginal Cost** curve must be increasing.*
- c) When should the company close?

16. Supply Curve in the SR

Short-run Supply Curve by a Price-taking Firm:

If the company is a price taker:

c) When should the company close?

In the short run, the firm must pay its fixed costs (even if the firm closes)

$$C_{SR} = \textit{Fixed Costs} + \textit{Variable Costs}_{SR}$$

If the firm offers q_2 at P_2 , the firm's profits will be:

$$\pi_{Open} = P_2 q_2 - \textit{Fixed Costs} - VC_{SR}$$

If the firm closes, it will have to pay the fixed costs, and its profits will be:

$$\pi_{Closed} = 0 - \textit{Fixed Costs}$$

16. Supply Curve in the SR

Short-run Supply Curve by a Price-taking Firm:

If the company is a price taker:

c) When should the company close?

To offer the good:

$$\pi_{Open} > \pi_{Closed}$$

$$Pq - \text{Fixed Costs} - VC_{SR} > -\text{Fixed Costs}$$

$$Pq > VC_{SR}$$

$$P > \frac{VC_{SR}}{q} = AVC_{SR}$$

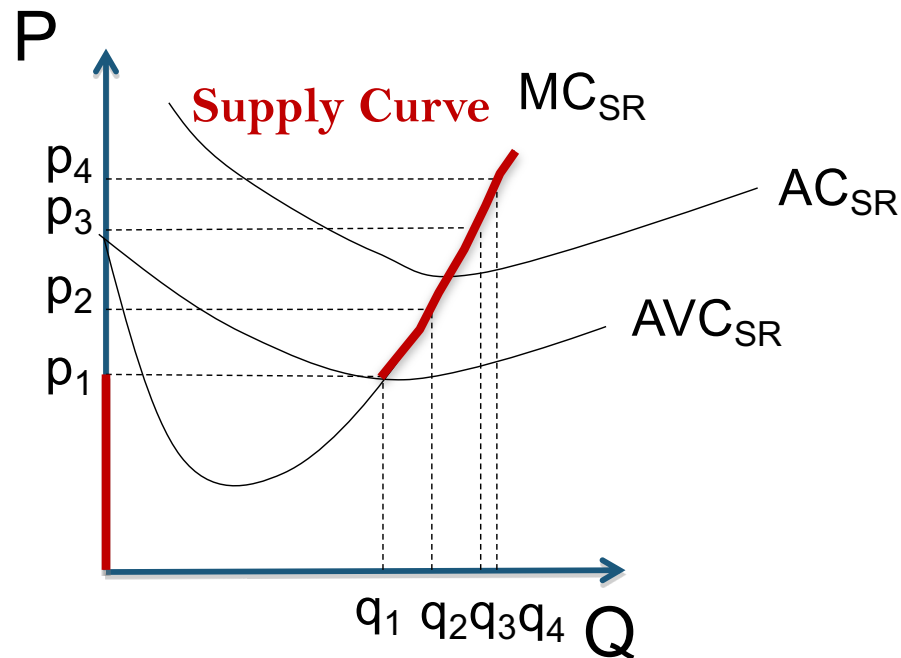
Attention: in the short run, the firm does not shut down when the price is less than the AC. It shuts down when the price is less than the AVC (although profits are negative the firm will prefer to supply this good, since it has to pay the fixed cost)

16. Supply Curve in the SR

Short-run Supply Curve by a Price-taking Firm:

If $P^* = P_3$, q_3 provides maximum profits, $\pi > 0$.

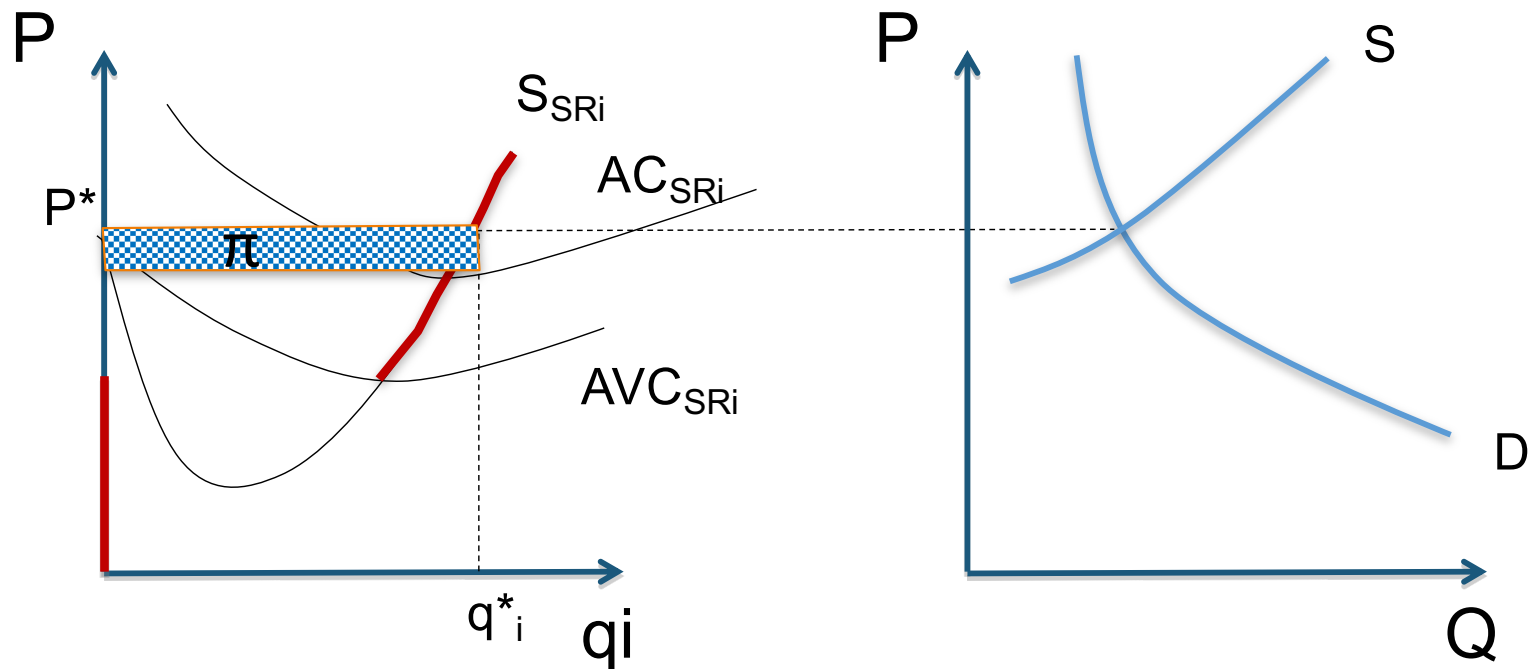
$$\text{Supply curve} = \begin{cases} MC = P & \text{for prices above } AVC_{SR} \\ 0 & \text{for prices below } AVC_{SR} \end{cases}$$



16. Supply Curve in the SR

Do not forget the bigger picture!

- The industry where this firm belongs has its own equilibrium where $S=D \rightarrow$ that's how we get the price P^*
- With that price we can find q_i^* and see the firm's benefit (π)





Author:
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(2021)

Activity 1

- Homework revisited: Let's do an example of the function cost. Imagine your firm faces a short-run cost function equal to:

$$C_{SR} = \bar{k}^2 + 16 \frac{Q^2}{\bar{k}}$$

- a) Find AC, MC, AVC in the short-run and graph them
- b) Find the supply curve for the short-run if you are a price-taker (*)**

Activity 1: Solution

Imagine your firm faces a short-run cost function equal to:

$$C_{SR} = \bar{k}^2 + 16 \frac{Q^2}{\bar{k}}$$

- a. Find AC, MC, AVC in the short-run and graph them

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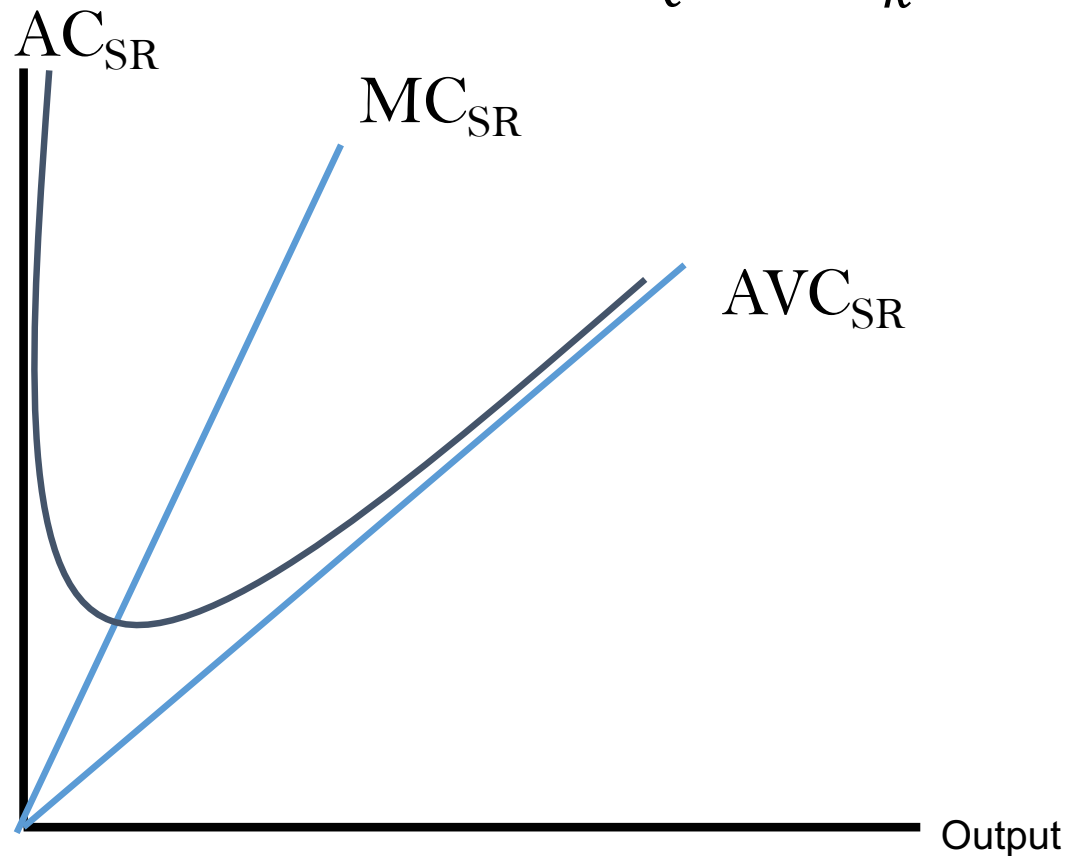
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Activity 1: Solution

$$AC_{SR} = \frac{C_{SR}}{Q} = \frac{\bar{k}^{-2}}{Q} + 16 \frac{Q}{\bar{k}}, MC_{SR} = \frac{\partial C_{SR}}{\partial Q} = 32 \frac{Q}{\bar{k}}$$

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Marginal Revenue

Producer Theory

15. Marginal Revenue

Elasticity of the demand curve is:

$$\varepsilon_{q,p} = \frac{dq / q}{dp / p} = \frac{dq}{dp} \frac{p}{q}$$

$$MR(q) = p + q \frac{dp}{dq} = p \left[1 + \frac{q}{p} \frac{dp}{dq} \right] = p \left[1 + \frac{1}{\varepsilon_{q,p}} \right]$$

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.